EARTHLENS AI

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EarthLens AI Technical Study Guide

- This study guide provides a comprehensive overview of the EarthLens AI platform, an end-to-end satellite intelligence system designed to democratize geospatial data analysis. It includes a review quiz, essay prompts, and a glossary of key technical terms.

Part 1: Review Quiz

- **Instructions:** Answer the following questions in 2–3 sentences based on the provided technical dossier.
  - **What is the "satellite data paradox" as defined in the platform's documentation?** The satellite data paradox refers to the fact that while humanity is in a "golden age" of abundant, free, and high-quality Earth observation data, the vast majority of people who need it—such as researchers and urban planners—cannot access or use it. This is due to structural barriers like software complexity, required domain expertise, and high infrastructure costs.

  - **Describe the four distinct architectural layers of the EarthLens AI platform.** The architecture consists of a Data Pipeline for API authentication and fetching, a Preprocessing Engine for normalization and masking, an Analysis and Intelligence layer for spectral computation and ML classification, and a Visualization Hub for rendering interactive maps and charts. Each layer is designed with a "separation of concerns" to remain modular and independently testable.

  - **How does the platform ensure functionality for users without active API credentials?** The platform includes an offline "synthetic mode" that generates realistic Sentinel-2 band data locally. This allows for complete development, testing, and exploration of the platform's features without the need for live API connections or credentials from GEE or Sentinel Hub.

  - **Explain the "dual-index masking approach" used in the Water Detection module.** To reduce false positives common in urban areas, EarthLens AI combines the Normalized Difference Water Index (NDWI) and the Modified Normalized Difference Water Index (MNDWI). A pixel is only classified as water if both indices return a value greater than zero, which suppresses urban signals that might otherwise be misidentified as water.

  - **What role does the Preprocessing Engine play in the data flow?** The Preprocessing Engine takes raw band arrays and performs vital tasks such as normalization to a 0–1 range, cloud pixel exclusion, and spatial resampling. For example, it resamples Sentinel-2 SWIR bands (native 20m resolution) to match the 10m resolution of optical bands to ensure a clean, aligned band stack for analysis.

- **Which machine learning models are utilized for land use classification, and what is their primary strength?** The platform uses Random Forest, Decision Tree, and Logistic Regression models. Random Forest provides the highest accuracy by handling non-linearity, Decision Trees offer fast and interpretable results, and Logistic Regression serves as a baseline model providing probabilistic outputs.

- **How is Normalized Difference Vegetation Index (NDVI) calculated and interpreted?** NDVI is calculated using the formula (NIR - Red) / (NIR + Red), exploiting the fact that healthy plants reflect near-infrared light strongly and red light weakly. Values ranging from 0.6 to 1.0 indicate dense vegetation like tropical forests, while values below 0.0 generally indicate water or snow.

- **What is the significance of the dNBR index in the Burn Area Detection module?** The dNBR (Difference Normalized Burn Ratio) measures fire severity by comparing pre-fire and post-fire NBR values. Because fire-affected areas lose chlorophyll (reducing NIR) and expose ash (increasing SWIR-2), the resulting dNBR value allows the platform to categorize damage from "Low Severity" to "High Severity" based on USGS standards.

- **Identify the three indices used to track urban expansion and specify which is most accurate.** The platform tracks urban areas using the Normalized Difference Built-Up Index (NDBI), the Urban Index (UI), and the Index-based Built-up Index (IBI). The IBI is considered the most accurate because it normalizes the built-up signal against vegetation and water signals, effectively extracting urban features.

- **What is Change Vector Analysis (CVA), and how does it improve change detection?** CVA is a method that captures both the magnitude and direction of spectral change between two time periods. By analyzing the directional signatures in the NIR-Red feature space, the platform can differentiate between various types of land changes, such as deforestation, urbanization, and flooding.

Part 2: Answer Key

Question	Core Concept to Include
1	Abundance of data vs. lack of accessibility for non-experts.
2	Data Pipeline, Preprocessing, Analysis/Intelligence, Visualization Hub.
3	Offline synthetic mode/local band generator.
4	Combination of NDWI and MNDWI > 0 to filter urban noise.
5	Normalization (0-1), cloud masking, and spatial resampling (10m/20m alignment).
6	Random Forest (accuracy), Decision Tree (speed), Logistic Regression (baseline).
7	(B08 - B04) / (B08 + B04); healthy plants reflect NIR and absorb Red.
8	Difference between pre-fire and post-fire NBR; identifies canopy loss/severity.
9	NDBI, UI, and IBI; IBI is the most accurate for urban extraction.
10	Magnitude + direction of spectral change; differentiates change types (e.g., flood vs. forest loss).

Part 3: Essay Format Questions

- **Addressing the Data Paradox:** Discuss how EarthLens AI's design philosophy of "software simplicity" and "embedded domain knowledge" specifically targets the structural barriers currently preventing the widespread use of satellite imagery in policy-making and environmental research.

- **Architectural Modularity:** Analyze the benefits of the four-layer modular architecture used in EarthLens AI. How does this structure support "Source Agnosticism" and the ability for the platform to scale with a user's technical access level?

- **Spectral Intelligence vs. Visual Interpretation:** Compare the utility of raw imagery with the spectral indices used in EarthLens AI (such as NDVI, MNDWI, and NDBI). How does mathematical band manipulation reveal information

invisible to the human eye?

- **Machine Learning in Geospatial Contexts:** Evaluate the EarthLens AI machine learning pipeline. Discuss the importance of feature engineering (the 11-feature matrix) and why multiple models (Random Forest vs. Logistic Regression) are necessary for robust land cover classification.

- **Future Technological Integration:** Based on the Version 2.0 roadmap, assess how the integration of Synthetic Aperture Radar (SAR) data and Deep Learning (CNNs) could enhance the platform's current capabilities in disaster response and agricultural monitoring.

Part 4: Glossary of Key Terms

Term	Definition
CVA (Change Vector Analysis)	A technique used to detect change by calculating the magnitude and direction of spectral shifts between two dates.
dNBR (Difference Normalized Burn Ratio)	The difference between pre-fire and post-fire NBR used to determine fire severity and canopy loss.
EVI (Enhanced Vegetation Index)	An optimized vegetation index designed to enhance the vegetation signal with improved sensitivity in high biomass regions.
GEE (Google Earth Engine)	A planetary-scale geospatial analysis cloud platform used by EarthLens AI for heavy computation and data fetching.
IBI (Index-based Built-up Index)	An advanced urban index that distinguishes built-up areas by normalizing them against vegetation and water indices.
MNDWI (Modified Normalized Difference Water Index)	An index that uses Green and SWIR-1 bands to detect water while suppressing signals from built-up urban areas.
NDBI (Normalized Difference Built-Up Index)	A spectral index used to identify urban and built-up areas by exploiting the contrast between SWIR and NIR reflectance.
NDVI (Normalized Difference Vegetation Index)	A standard index for quantifying vegetation greenness and health based on the ratio of Near-Infrared and Red light.
NIR (Near-Infrared)	A portion of the electromagnetic spectrum (approx. 842 nm) highly reflected by healthy vegetation.
SAR (Synthetic Aperture Radar)	A form of radar used to create 2D images or 3D reconstructions of objects; planned for future flood mapping modules.
Source Agnosticism	A design principle where the core analysis engine can process data regardless of whether it originates from GEE, USGS, or synthetic sources.
SWIR (Short-Wave Infrared)	Wavelengths used to detect moisture in vegetation and soil, as well as to differentiate between different types of land cover.

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